

Table S1

Symbol	Name/Definition	Normal Steady-State value/units
$A_{\text{aldo-cd}}$	fitting constant for aldosterone effect on collecting duct reabsorption	0.74
$A_{\text{aldo-dt}}$	fitting constant for aldosterone effect on distal tubule sodium reabsorption	0.24
$A_{\text{AT1-AA}}$	fitting constant for AT1-bound AngII effect on afferent arteriole resistance	0.8
$A_{\text{AT1-EA}}$	fitting constant for AT1-bound AngII effect on efferent arteriole resistance	0.925
$A_{\text{AT1-preglom}}$	fitting constant for AT1-bound AngII effect on preglomerular arteriole resistance	0.8
$A_{\text{at1-renin}}$	fitting constant for AT1-bound AngII feedback on renin secretion	0.0102
$A_{\text{md-renin}}$	fitting constant for macula densa feedback on renin secretion	0.2262
$\text{AT1-bound_AngII}_{\text{EQ}}$	equilibrium level of AT1-bound AngII	15.8 pg/ml
$B_{\text{aldo-cd}}$	fitting constant for aldosterone effect on collecting duct reabsorption	0.06
$B_{\text{aldo-dt}}$	fitting constant for aldosterone effect on distal tubule sodium reabsorption	0.3
$B_{\text{AT1-AA}}$	fitting constant for AT1-bound AngII effect on afferent arteriole resistance	0.055
$B_{\text{AT1-EA}}$	fitting constant for AT1-bound AngII effect on efferent arteriole resistance	0.05
$B_{\text{AT1-preglom}}$	fitting constant for AT1-bound AngII effect on preglomerular arteriole resistance	0.055
$B_{\text{at1-renin}}$	fitting constant for AT1-bound AngII feedback on renin secretion	0.95
$B_{\text{md-renin}}$	fitting constant for macula densa feedback on renin secretion	28.04
C_{ACE}	rate of conversion of AngI to AngII by ACE	54.1 /h
C_{ACE2}	rate of conversion of AngII to Ang(1-7) by ACE2	2.4 /h

$C_{\text{AngII-AngIV}}$	rate of conversion of AngII to AngIV	23.5 /h
C_{AT1}	rate of AngII binding to AT1 receptor	11.8 /h
$C_{\text{AT1-AA}}$	fitting constant for AT1-bound AngII effect on afferent arteriole resistance	0.185
$C_{\text{AT1-EA}}$	fitting constant for AT1-bound AngII effect on efferent arteriole resistance	0.17
$C_{\text{AT1-preglom}}$	fitting constant for AT1-bound AngII effect on preglomerular arteriole resistance	0.185
C_{AT2}	rate of AngII binding to AT2 receptor	3.9 /h
C_{chymase}	rate of conversion of AngI to AngII by chymase	1.1 /h
$C_{\text{GP_autoreg}}$	gain of glomerular pressure autoregulatory feedback	5
$C_{\text{md-renin}}$	fitting constant for macula densa feedback on renin secretion	2
C_{nep}	rate of conversion of AngI to Ang(1-7) by neprilysin	1.1 /h
h_{agt}	angiotensinogen half life	10 hr
$h_{\text{ANG(1-7)}}$	half life of Ang(1-7)	30 min
h_{ANGI}	half life of AngI	0.5 min
h_{AngII}	half life of AngII	0.66 min
h_{ANGIV}	half life of AngIV	0.5 min
h_{AT1}	half life of AT1-bound AngII	12 min
h_{AT2}	half life of AT2-bound AngII	12 min
h_{renin}	half life of circulating renin	12 min
k_{agt}	rate of angiotensinogen production	34,620 fmol/ml/h
K_f	single nephron hydraulic conductance	3.905 nl/min/mmHg
N_{nephrons}	typical number of nephrons in a healthy individual	2×10^6

N_{rs}	nominal renin secretion rate	97 ng/ml/hr
$\text{Offset}_{\text{md-renin}}$	fitting constant for macula densa feedback on renin secretion	5
P_B	Bowman hydrostatic pressure	18 mmHg
P_{gh_nom}	nominal glomerular hydrostatic pressure	60 mmHg
P_{go}	glomerular osmotic pressure	28 mmHg
$\text{PRC}(0)$	equilibrium plasma renin concentration	15 pg/ml
r_{aa_0}	nominal single afferent arteriole resistance	6×10^6 mmHg*min/L
r_{ea_0}	nominal single efferent arteriole resistance	10^7 mmHg*min/L
$R_{\text{preglom_0}}$	nominal resistance of preglomerular arteries	15 mmHg*min/L
$X_{\text{md-renin}}$	fitting constant for macula densa feedback on renin secretion	1.15
$X_{\text{prc-pra}}$	equilibrium ratio of PRA to PRC	61 fmol/hr/pg